Universidade de São Paulo
Instituto de Astronomia, Geofísica e Ciências Atmosféricas

Air pollution modeling in the Estado de São Paulo using a bottom-up emissions inventory

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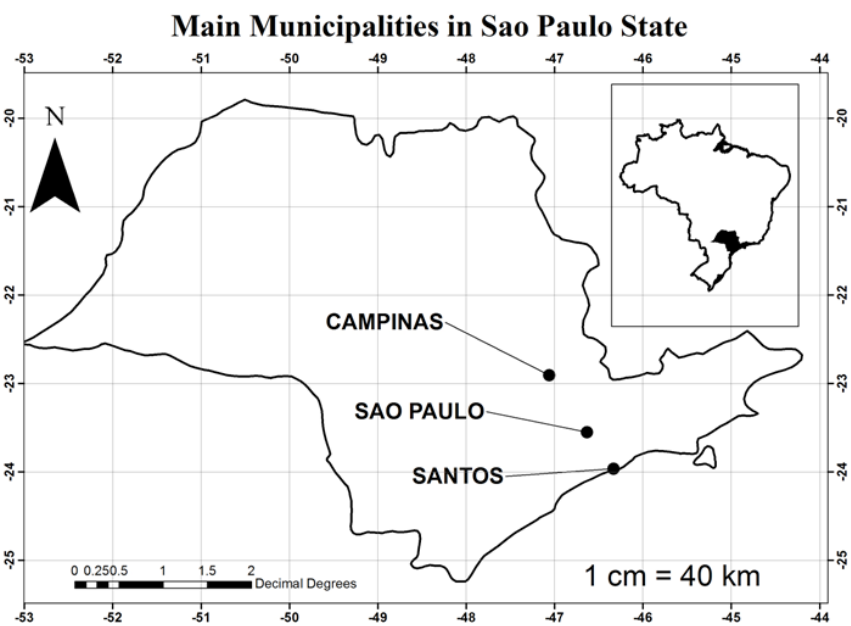
2/10/14

1 Background

- Metropolitan Region of Sao Paulo, Campinas and Santos.
- 20 million inhabitants.
- In spring time, Tropospheric Ozone levels exceed the 160 µg/m3 hourly Brazilian Standards.
- Precursors of Ozone NOx, VOC and solar radiation.



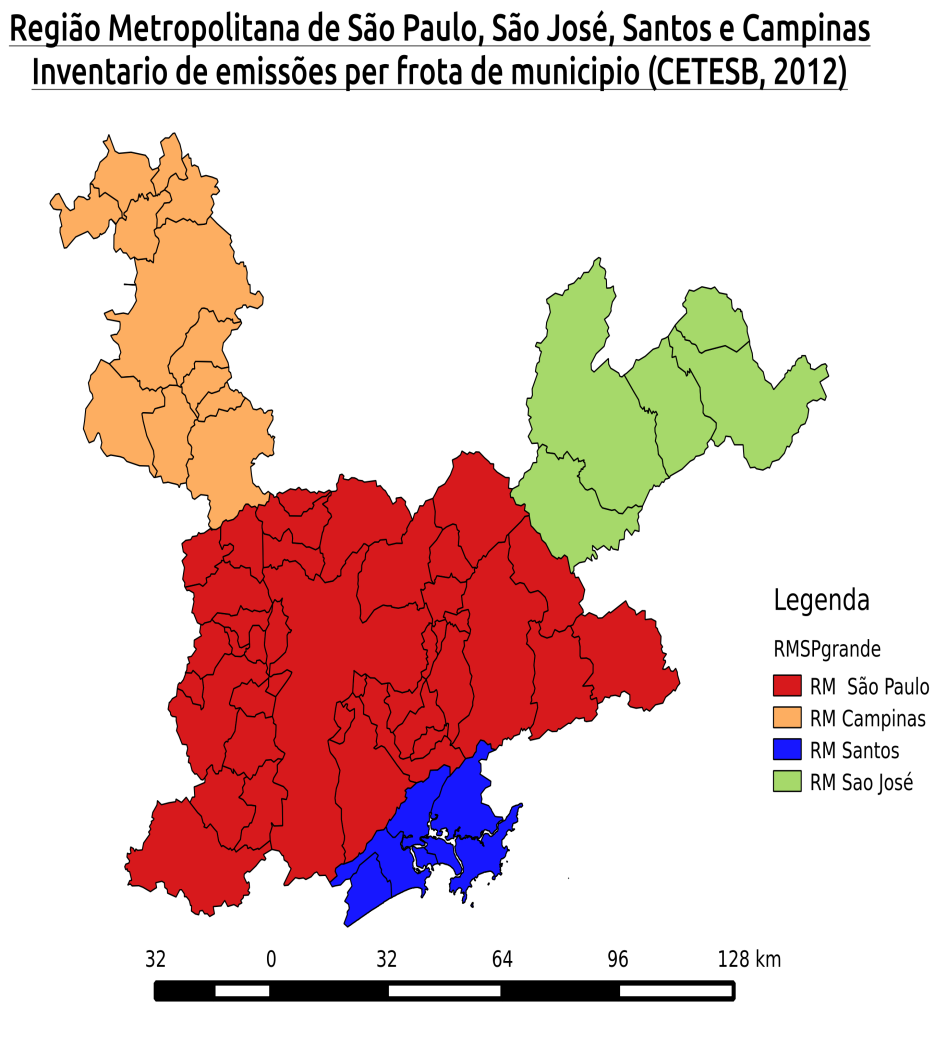
- Expanded Area: Sao Paulo, Campinas, Santos and more

1 Background

$$E = FE * \underbrace{VF * TV * L}_{\text{Activity Level}}$$

- E: Emissions
- FE: Emissions factors
- VF: Vehicular flux per link of road
- TV: Type of Vehicle (technology, standard)
- L: Length of the link of road

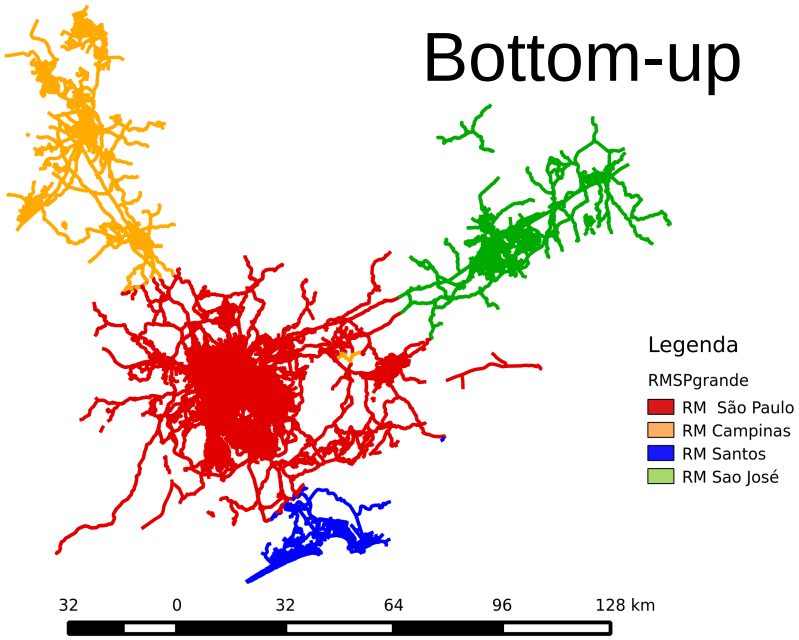
Where are the cars?



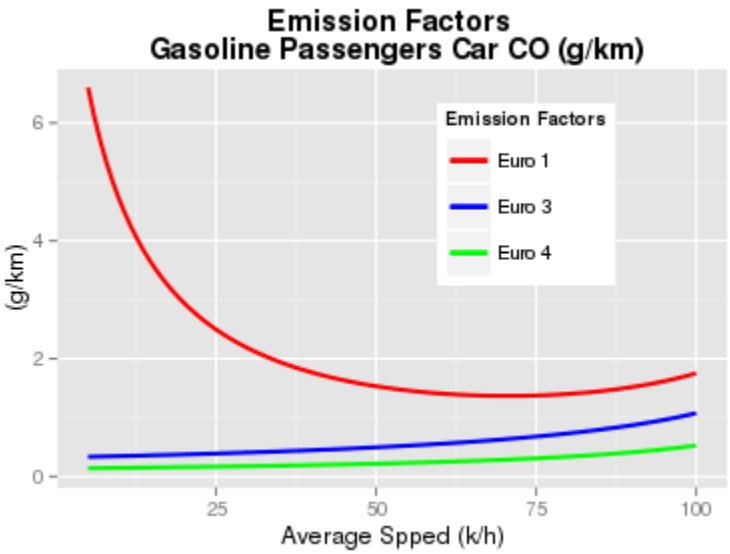
Top-Down

But in reality...

Região Metropolitana de São Paulo, São José, Santos e Campinas
Inventario de emissões rua per rua



Dynamics of vehicular flux should be incorporated to emissions models



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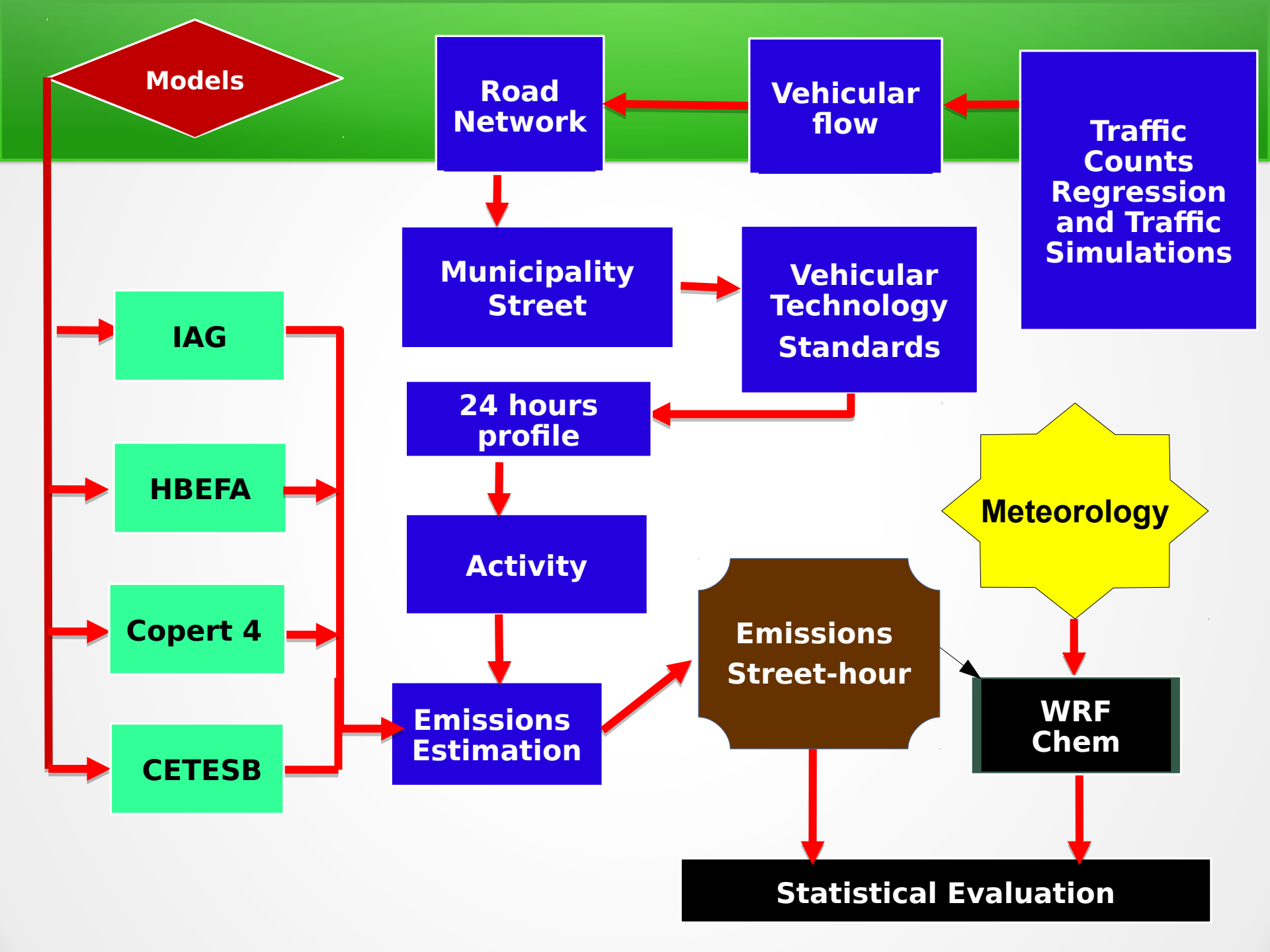
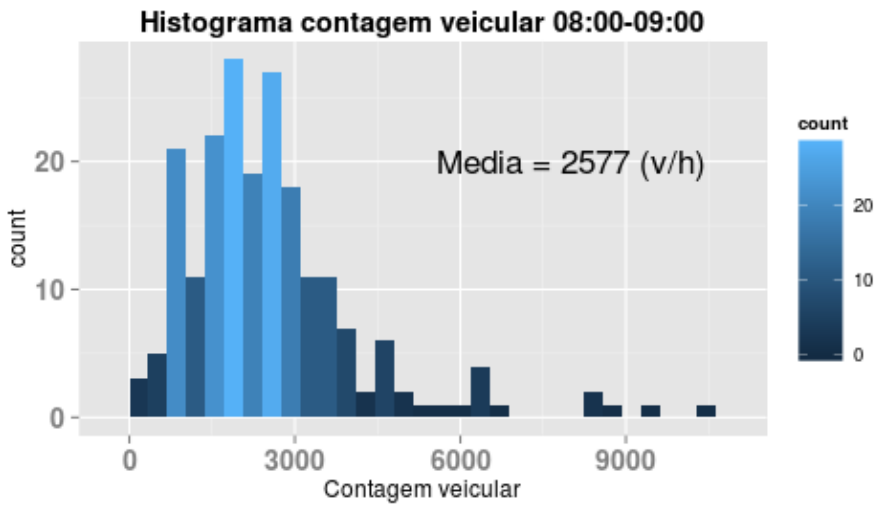
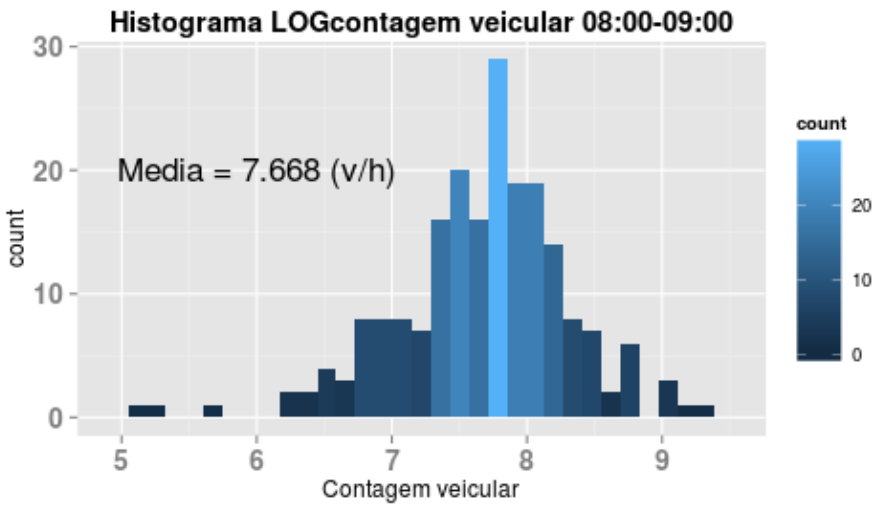

Models
Road Network
Vehicular flow
Traffic Counts Regression and Traffic Simulations
Municipality Street
Vehicular Technology Standards
IAG
24 hours profile
HBEFA
Meteorology
Activity
Copert 4
Emissions Street-hour
Emissions Estimation
WRF Chem
CETESB
Statistical Evaluation

Metodos





206 points for counting

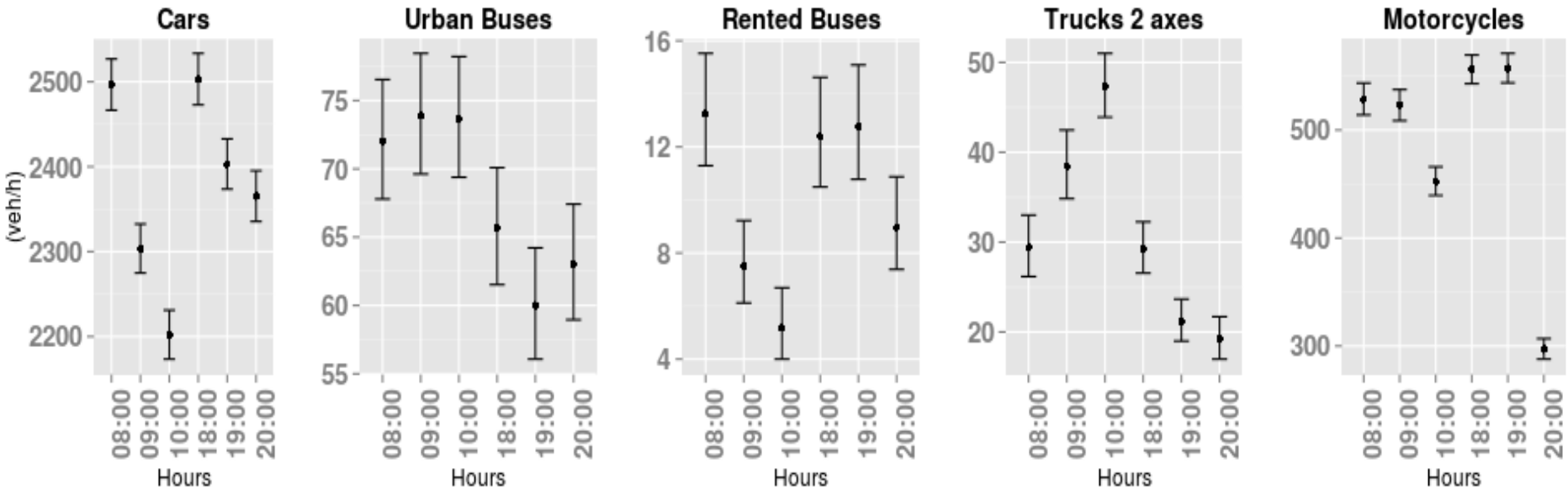
Regressions:

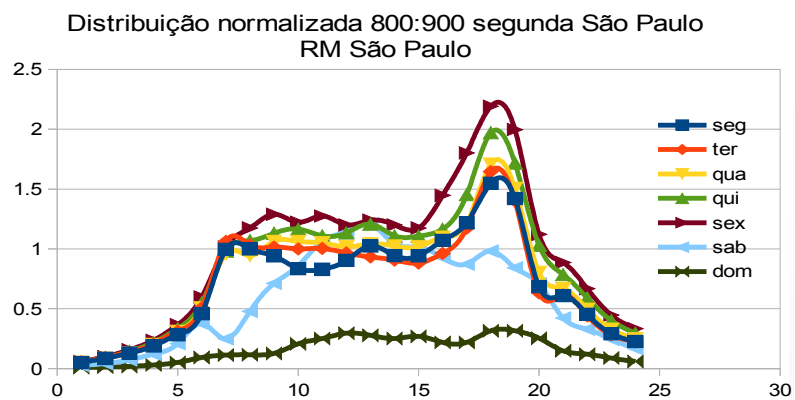
- Distribution Poisson
- Zero Inflated Negative Binomial
- Normal

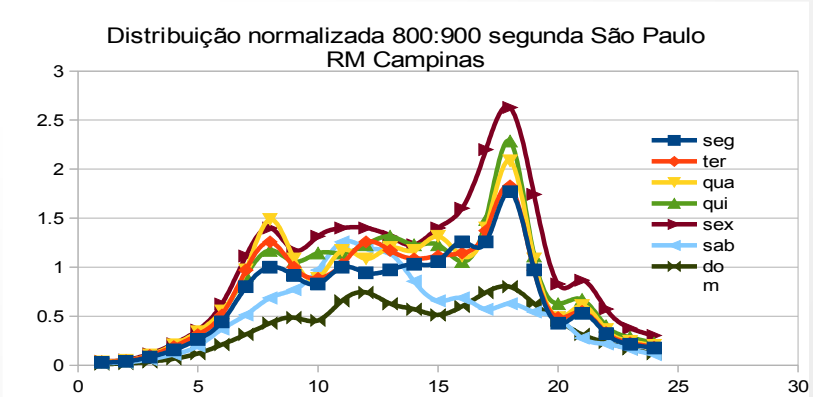
Model

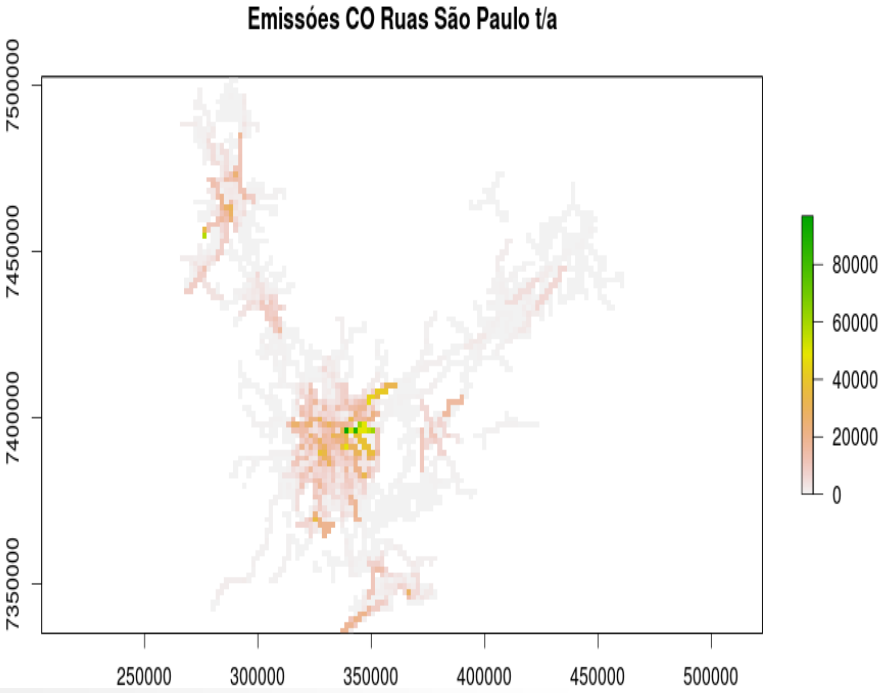
$$\log(veh) = \beta_0 + \beta_1 * TR + \varepsilon$$

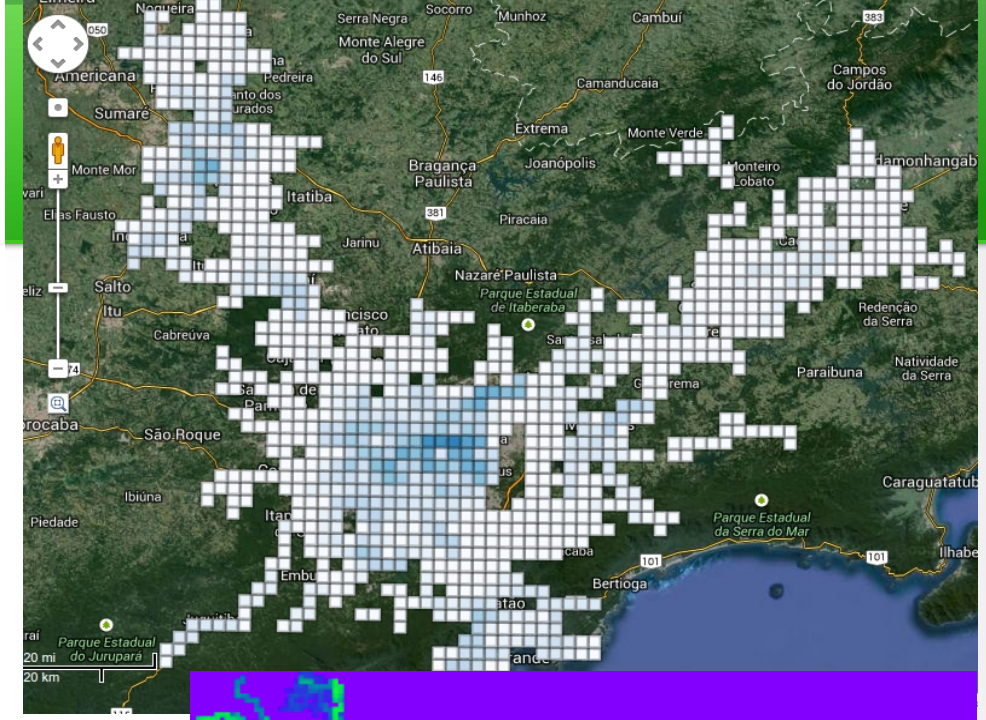
βo: intercept; β1: estimate; TR: Type of Street; ε: Error

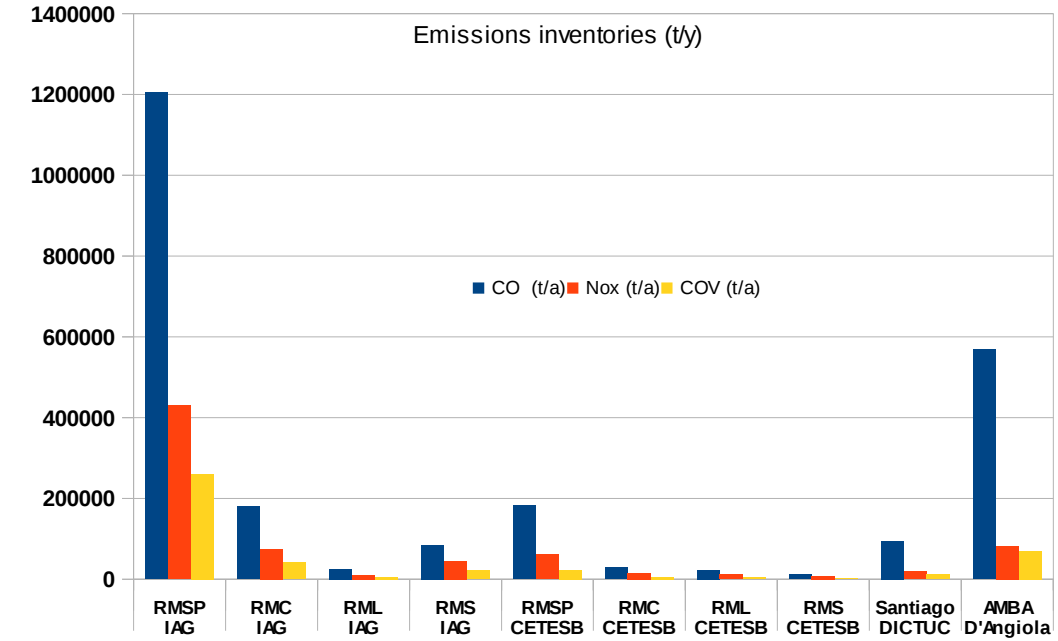
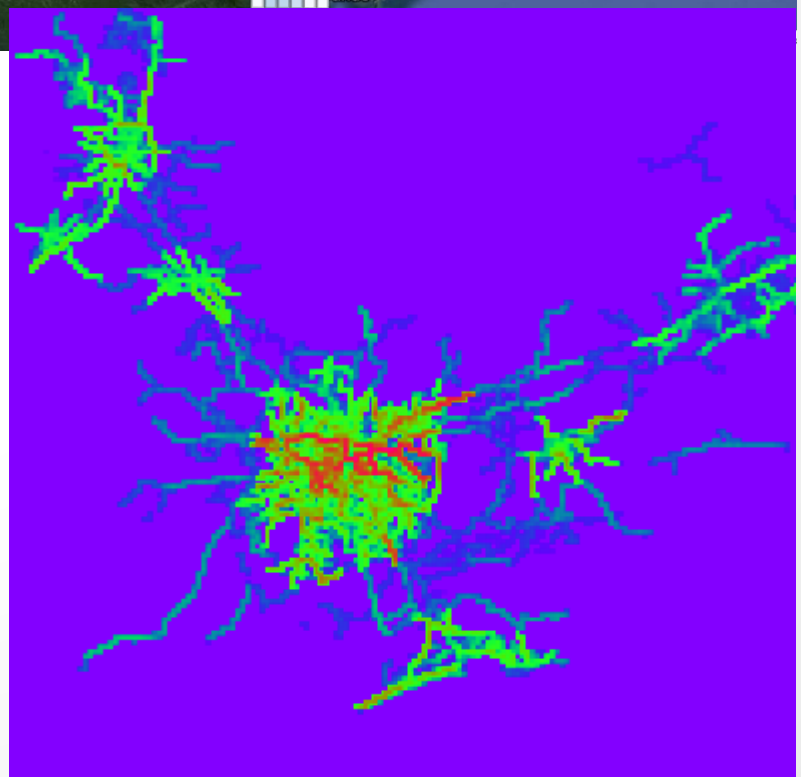







Emissões CO Ruas São Paulo t/a
7500000
7450000
7400000
7350000
250000
300000
350000
400000
450000
500000
80000
60000
40000
20000
0


Americana
Sumaré
Monte Mor
Itatiba
Bragança Paulista
Atibaia
Nazaré Paulista
Parque Estadual de Itaberaba
Salto
Itu
Cabreúva
São Roque
Ibiúna
Piedade
Parque Estadual do Jurupará
Parque Estadual da Serra do Mar
Bertioga
Paraibuna
Campos do Jordão
Extrema
Joanópolis
Piracaia
Jarinu
Monte Verde
Camanducaia
Cambuí
Munhoz
Socorro
Serra Negra
Monte Alegre do Sul
Pedreira
Redenção da Serra
Natividade da Serra
Caraguatatuba
Ilhabela
20 mi
20 km


Emissions inventories (t/y)
1400000
1200000
1000000
800000
600000
400000
200000
0
CO (t/a)
Nox (t/a)
COV (t/a)
RMSP IAG
RMC IAG
RML IAG
RMS IAG
RMSP CETESB
RMC CETESB
RML CETESB
RMS CETESB
Santiago DICTUC
AMBA D'Angiola